Constructive Dynamical Axioms for Quantum Field Theory

Version VII: Core Axioms, Specialization Axes, Spectral Type, Scope Boundaries, and Detailed Framework Correspondences

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Abstract

The standard axiomatic frameworks for quantum field theory — Wightman, Osterwalder–Schrader, and Haag–Kastler / locally covariant AQFT — describe the output of a quantum field theory rather than the constructive dynamical system that produces it. The present paper proposes a framework whose primitive object is the constructive dynamical system itself, organized into five core axioms shared by all constructive dynamical QFTs with propagating local degrees of freedom — geometric locality, finite resolution, finite propagation, local dynamics with additive decomposition, and positivity/stability — together with four specialization axes for internal symmetry, statistics, interaction type, and spectral type. The framework is designed so that the standard axiomatic frameworks reappear as specialization branches under explicitly stated supplementary hypotheses: in the flat stationary sector, the core axioms together with branch data and transfer-matrix hypotheses yield the Osterwalder–Schrader axioms OS0–OS4 and hence, by reconstruction, the Wightman axioms; on general globally hyperbolic manifolds, the same core-plus-branch structure induces a locally covariant net of physical $*$-algebras. The paper isolates upstream structural conditions and maps them in detail to the standard frameworks, separating universal framework statements from the additional analytic hypotheses needed for clustering, transfer-matrix factorization, hyperbolicity, and Hadamard regularity. Its object is the abstract constructive architecture itself.

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1 Introduction

1.1 Problem Statement

The mathematical foundations of quantum field theory rest on three standard axiomatic frameworks, each native to a particular geometric setting.

The Wightman axioms [?, ?] take as primitive operator-valued tempered distributions on Minkowski space acting on a Hilbert space carrying a positive-energy representation of the Poincaré group. They encode locality, relativistic covariance, and the spectral condition, but they are stated in terms of the output of the theory rather than in terms of a constructive dynamical system that generates that output.

The Osterwalder–Schrader axioms [?, ?] take as primitive the Schwinger functions on flat Euclidean space and impose temperedness, Euclidean covariance, reflection positivity, clustering, and graded symmetry. They are the gateway from Euclidean construction to Minkowskian QFT, but again they operate downstream, at the level of correlation functions already given as output data.

The Haag–Kastler axioms and their locally covariant generalization [?, ?] take as primitive a net, or functor, of local $*$-algebras over globally hyperbolic spacetimes. These frameworks are manifold-native and conceptually powerful, but they organize observables rather than specifying the constructive dynamics that produces them.

These frameworks are complementary rather than competing. The problem is that they are fragmented. There is no widely used upstream axiom system whose primitive object is the constructive dynamical system itself and which is explicitly designed to specialize into all three standard frameworks.

1.2 Main Idea

The framework has two layers.

- **Core axioms**: structural requirements shared by all constructive QFTs, independent of field content or phase.
- **Specialization axes**: branch conditions that classify theory type and selected phase.

The five core axioms are:

1. Geometric locality;
2. Finite resolution;
3. Finite propagation;
4. Local dynamics with additive decomposition;
5. Positivity and stability.

The specialization data are organized along four axes:

Axis A: Internal symmetry (gauge, global, or discrete gauge);
Axis B: Statistics (bosonic, fermionic, mixed, or supersymmetric mixed);
Axis C: Interaction type (interacting or free);
Axis D: Spectral type (massive/gapped or critical/conformal).

The distinction between a gapped theory and a critical/conformal theory is not built into the core axioms; it is selected by parameter choice or phase selection inside the same constructive architecture.

1.3 Nature of This Paper

This is a framework paper. The core axioms isolate upstream structural conditions. The specialization theorems show that, under explicitly stated supplementary hypotheses, the standard frameworks reappear as abstract branches. The paper does not claim that the core axioms automatically produce those hypotheses. It sharply separates three layers of verification:

1. **Structural verification**: the core axioms and branch data;
2. **Specialization verification**: the flat-stationary or manifold-covariant branch theorem;
3. **Supplementary analytic hypotheses**: transfer-matrix factorization, spectral gap or mixing input, hyperbolicity for the time-slice property, Hadamard regularity, and related conditions.

The paper cleanly identifies which upstream input produces which downstream property and which steps require additional analytic hypotheses.

1.4 Main Theorems

The paper is organized around three results, stated here in a form that makes the additional hypotheses explicit.

Theorem 1.1 (Flat Stationary Specialization)**.** *In the flat stationary sector* $(M, g) = (\mathbb{R}^4, \delta)$*, a system satisfying Core Axioms 1–5 together with branch choices on Axes A, B, and C produces Schwinger functions satisfying:*

(i) OS0 from Core Axiom 2;

(ii) OS1 from Core Axiom 4 in the flat Euclidean sector, under the hypothesis that the equilibrium measure is the unique $E(4)$*-invariant measure associated with the action;*

(iii) OS2 from Core Axioms 4 and 5, under the transfer-matrix hypotheses of Proposition **??***;*

(iv) OS4 from the chosen branch on Axis B.

If, in addition, the translation-mixing estimate (**??**) *holds for the translation-stable observable algebra generating the Schwinger functions, then OS3 follows. Axis D refines the expected clustering rate: gapped phases (D1) yield exponential decay, while critical/conformal phases (D2) yield polynomial decay. OS reconstruction then yields Wightman functions satisfying W0–W4.*

Proof. This is the summary form of Theorem **??**. The structural derivations of OS0, OS1, OS2, and OS4 are proved there, and OS3 is supplied by Proposition **??**. The final reconstruction step is the standard Osterwalder–Schrader reconstruction theorem [**?**, **?**]. □

Theorem 1.2 (Manifold-Covariant Specialization)**.** *On a general globally hyperbolic Lorentzian manifold* (M, g)*, a system satisfying Core Axioms 1–5, a symmetry branch on Axis A, and geometric naturality of the generating structure in the sense of Remark* **??** *induces a locally covariant net of physical* $*$*-algebras satisfying isotony, locality, and local covariance. In the gauge branches (A1 or A3), the physical net is the invariant subalgebra; in the global-symmetry branch (A2), the full local algebras are physical and the global symmetry acts by automorphisms. If the field equations of the theory are normally hyperbolic or of Dirac type, then the time-slice property holds. If the physical states are Hadamard, then the microlocal spectrum condition holds.*

Proof. This is the summary form of Theorem **??**. The net structure, covariance morphisms, and the two additional conclusions are proved there from Core Axioms 1–5 together with Remark **??**. □

Theorem 1.3 (Structural Separation). *The five core axioms together with the four specialization axes determine a universal separation between structural outputs and supplementary analytic inputs. In particular, temperedness, algebraic locality, positivity, and branch-dependent grading are fixed at the framework level, whereas reflection positivity beyond hyperplane decomposition, clustering, time-slice, and microlocal spectrum statements require the additional hypotheses made explicit in Sections* **??** *and* **??**.

Proof. The separation is obtained by reading the later derivations result by result. Core Axiom **??** yields OS0 directly; Core Axiom **??** together with Axis B determines the algebraic locality and grading data; Core Axiom **??** gives the positive semigroup and hence the GNS positivity input; and Axis B fixes graded symmetry. By contrast, Proposition **??** shows that OS2 requires, in addition to the core axioms, explicit transfer-matrix hypotheses. Proposition **??** shows that OS3 requires the translation-mixing hypothesis (**??**). Theorem **??** shows that local covariance uses the extra geometric naturality stated in Remark **??**, while time-slice and microlocal spectrum require respectively hyperbolicity and Hadamard regularity. This is exactly the claimed structural/analytic separation. □

1.5 Verification Interface and Frozen Hypothesis Packages

To make the framework reusable in downstream realization papers, the flat-sector verification route is packaged here as a theorem library with named hypothesis bundles. The point is not to change the logical architecture already established above, but to freeze it into a citation-ready interface: one verifies a structural package, then a small number of named supplementary packages, and then invokes a single theorem. Downstream realization papers should therefore be read in three blocks:

(i) **Implementation data:** the concrete realization tuple (state space, symmetry maps, observable algebras, transfer maps, semigroup data);

(ii) **Verified laws:** the package obligations H_{struct}, RP, CL, MG, TS, H;

(iii) **Returned outputs:** OS0–OS4, W0–W4, and the gap conclusion allowed by the master interface theorem.

Definition 1.4 (Structural flat package). The *structural flat package* H_{struct} consists of:

(S1) Core Axioms 1–5 in the flat stationary sector $(\mathbb{R}^4, \delta)$;

(S2) a branch choice on Axes A, B, and C;

(S3) a chosen $E(4)$-invariant equilibrium state in the sense used for OS1;

(S4) the positive-time physical algebra and its centered subspace as specified in Definition **??**.

Definition 1.5 (Structural manifold package). The *structural manifold package* H_{man} consists of:

(M1) Core Axioms 1–5 on a globally hyperbolic Lorentzian manifold;

(M2) a choice on Axis A fixing the physical algebra;

(M3) geometric naturality in the sense of Remark **??**.

Definition 1.6 (Frozen supplementary packages). In the flat stationary sector, the reusable supplementary packages are the following.

Package RP (reflection positivity / transfer). RP consists of the hypotheses of Proposition **??**:

(RP1) hyperplane decomposition as in Lemma **??**;

(RP2) a reflection-invariant hyperplane-factorizing Euclidean measure associated with the generating functional;

(RP3) implementation of Euclidean time evolution by the positivity-preserving semigroup of Core Axiom **??**.

Package CL (clustering). CL consists of:

(CL1) a unital translation-stable observable algebra $\mathcal{B}^{\mathrm{phys}}$ from which the Schwinger functions are generated;

(CL2) the translation-mixing estimate (**??**) on $\mathcal{B}^{\mathrm{phys}}$.

Package MG (mass-gap transfer). MG consists of:

(MG1) a linear subspace $\mathcal{C}_+^{\mathrm{phys},0} \subset \mathcal{A}_+^{\mathrm{phys},0}$ of centered positive-time observables whose OS images are well-defined in the reconstructed Hilbert space;

(MG2) a semigroup estimate on that class: there exist $\lambda_* > 0$ and $C_{\mathrm{MG}} \geq 1$ such that for all $F \in \mathcal{C}_+^{\mathrm{phys},0}$ and $t \geq 0$,

$$\|e^{-tH}\widehat{F}\|_{\mathcal{H}} \leq C_{\mathrm{MG}} e^{-\lambda_* t} \|\widehat{F}\|_{\mathcal{H}};$$

(MG3) density of the OS images $\{\widehat{F} : F \in \mathcal{C}_+^{\mathrm{phys},0}\}$ in $\Omega^\perp$;

(MG4) compatibility of the class with the reconstructed physical sector, so that the above estimate is a statement about the same Hamiltonian that appears in OS reconstruction.

Package TS (time-slice). Relative to the structural manifold package H_{man}, TS consists of:

(TS1) the local algebra is generated by on-shell fields modulo the equations of motion;

(TS2) normal hyperbolicity or Dirac-type hyperbolicity of those equations.

Package H (Hadamard / microlocal spectrum). Relative to the structural manifold package H_{man}, H consists of:

(H1) the requirement that the physical states under consideration are Hadamard.

Remark 1.7 (Package schemas are local contracts). Definitions **??**–**??** should be read as typed schemas rather than as narrative summaries. A downstream realization implements the framework by:

(i) specifying concrete data for each slot consumed by a package;

(ii) verifying the corresponding law clauses item-by-item;

(iii) invoking the master interface theorem only after those clauses have been checked.

In particular, package names such as RP, CL, and MG are abbreviations for explicit finite lists of obligations, not black-box citations.

Remark 1.8 (Imported standard results versus framework results). The verification API separates three layers.

(i) **Framework results proved here:** Proposition **??**, Proposition **??**, Theorem **??**, Proposition **??**, Proposition **??**, and Theorem **??**.

(ii) **Imported standard results:** the classical Osterwalder–Schrader reconstruction theorem once OS0–OS4 are verified, the spectral theorem for the reconstructed Hamiltonian, and the standard hyperbolic/Hadamard implications in the manifold branch.

(iii) **Realization-specific work:** verification that a given model satisfies the packages RP, CL, MG, TS, and H.

Theorem 1.9 (Master verification interface)**.** *Assume the structural flat package H_{struct} of Definition* **??**.

(a) If Package RP holds, the interface returns OS2.

(b) If Package CL holds, the interface returns OS3.

(c) If both Package RP and Package CL hold, the interface returns OS0–OS4, hence the standard Osterwalder–Schrader reconstruction theorem yields Wightman functions satisfying W0–W4.

(d) If, in addition, Package MG holds, the interface returns the reconstructed Hamiltonian satisfies

$$\mathrm{spec}(H) \cap (0, \lambda_*) = \varnothing.$$

Proof. Under H_{struct}, OS0, OS1, and OS4 are supplied by Theorem **??**. Package RP is exactly the hypothesis set of Proposition **??**, which proves OS2. Package CL is exactly the hypothesis set of Proposition **??**, which proves OS3. Once OS0–OS4 are available, the standard Osterwalder–Schrader reconstruction theorem yields W0–W4. The mass-gap conclusion is Theorem **??**. □

Remark 1.10 (Return type of the flat verification interface)*.* Theorem **??** should be read as returning a structured output object, not merely a yes/no verdict. Once a realization has supplied implementation data and verified the package laws, the interface returns:

(i) Schwinger functionals satisfying the requested OS clauses;

(ii) when OS0–OS4 are all available, the standard reconstructed Hilbert space, vacuum vector, and positive Hamiltonian from classical Osterwalder–Schrader reconstruction;

(iii) when Package MG is also available, the spectral-gap conclusion for that same reconstructed Hamiltonian.

Downstream realization papers may therefore treat the framework theorem as a contract that consumes verified package data and returns mathematically typed outputs.

Theorem 1.11 (Master manifold verification interface)**.** *Assume the structural manifold package H_{man} of Definition* **??**.

(a) Isotony, locality, physical-algebra selection, and local covariance follow.

(b) If, in addition, Package TS holds, then the time-slice property holds.

(c) If, in addition, Package H holds, then the microlocal spectrum condition holds for the physical states under consideration.

Proof. This is Theorem **??** rewritten in package form relative to H_{man}. □

Corollary 1.12 (Gauge-invariant sector interface)**.** *In the gauge branches A1 or A3, the preceding interface applies after replacing the ambient observable algebra by the gauge-invariant physical algebra $\mathcal{A}_M^{\mathrm{phys}}(O) = \mathcal{A}_M(O)^G$ or $\mathcal{A}_M^{\mathrm{phys}}(O) = \mathcal{A}_M(O)^{G_{\mathrm{disc}}}$. If that gauge-invariant sector satisfies Packages RP and CL, then OS0–OS4 and W0–W4 hold on the gauge-invariant reconstructed sector. If, further, Package MG holds there, then the reconstructed gauge-invariant Hamiltonian has no spectrum in* $(0, \lambda_*)$.

Proof. Axis A fixes the physical algebra. All statements in Theorem **??** are formulated directly on that physical algebra, so the gauge-invariant specialization is immediate. □

Remark 1.13 (Dependency diagram with theorem labels). The framework's verification flow can be read as the diagram

$$H_{\text{struct}} \overset{\text{Theorem ??}}{\Longrightarrow} \text{OS0}, \text{OS1}, \text{OS4}, \qquad H_{\text{struct}} + \text{RP} \overset{\text{Proposition ??}}{\Longrightarrow} \text{OS2},$$

$$H_{\text{struct}} + \text{CL} \overset{\text{Proposition ??}}{\Longrightarrow} \text{OS3}, \qquad H_{\text{struct}} + \text{RP} + \text{CL} \overset{\text{Theorem ??}}{\Longrightarrow} \text{OS0–OS4} \Rightarrow \text{W0–W4},$$

$$H_{\text{struct}} + \text{RP} + \text{CL} + \text{MG} \overset{\text{Theorem ??}}{\Longrightarrow} \operatorname{spec}(H) \cap (0, \lambda_*) = \varnothing,$$

$$H_{\text{man}} \overset{\text{Theorem ??}}{\Longrightarrow} \text{HK/LC}, \qquad + \text{TS} \Rightarrow \text{time-slice}, \quad + \text{H} \Rightarrow \text{microlocal spectrum}.$$

This is the theorem-level checklist that later realization papers are meant to discharge.

1.6 Relation to Recent Constructive QFT

The present framework should be positioned relative to the active frontiers of rigorous constructive QFT. Hairer's theory of regularity structures [?] and the paracontrolled distributions of Gubinelli, Imkeller, and Perkowski [?] have made it possible to construct solutions to singular stochastic PDEs — including the dynamical Φ^4_3 model — that were previously beyond rigorous reach. The Balaban–Dimock–Imbrie–Jaffe program [?, ?] pursues constructive Yang–Mills theory through multiscale cluster expansions.

These programs develop concrete constructions. The present framework is not intended to compete with them but to provide an organizational layer above them: a common set of structural conditions and specialization rules that can be used to compare different constructive approaches without committing the framework itself to a particular one.

1.7 Roadmap

Section ?? develops the detailed correspondence between the present framework and the three standard axiomatic frameworks. Section ?? states the core axioms, with particular attention to the reformulation of Core Axiom 4 and the causal-structure role of Core Axiom 3. Section ?? states the specialization axes. Section ?? proves the specialization theorems, with an explicit treatment of the upstream dynamical chain leading to OS3. Section ?? discusses compatibility, non-redundancy, and core separating examples. Section ?? explains in what sense the framework is constructive and clarifies the role of renormalization and scaling. Section ?? treats supersymmetry and the TQFT scope boundary. Section ?? records limitations and future refinements. Section ?? concludes.

2 The Standard Frameworks and Their Upstream Sources

This section develops the detailed correspondence between the five core axioms (plus specialization data) and each of the three standard axiomatic frameworks. For each standard axiom, we identify: which core axiom provides the primary structural input; which additional hypotheses, if any, are needed; and where the hard analytic work resides. This makes explicit the claim that the present framework is genuinely upstream of the existing ones.

2.1 Wightman Axioms

The Wightman axioms [?, ?] take as primitive a separable Hilbert space $\mathcal{H}$, a vacuum vector $|\Omega\rangle$, a unitary strongly continuous representation of the proper orthochronous Poincaré group, and a collection of operator-valued tempered distributions obeying covariance, locality, the spectral condition, and cyclicity of the vacuum. In the present framework, none of these are primitive; they are all derived.

W0: Hilbert space, vacuum, Poincaré representation, spectral condition. Core Axiom 5 provides a positivity-preserving semigroup with self-adjoint generator $H \geq 0$ and an equilibrium state ω. The GNS construction on ω produces $(\mathcal{H}, \pi, |\Omega\rangle)$. Uniqueness of the vacuum follows from OS3, which requires the translation-mixing hypothesis of Proposition **??**. The Poincaré representation arises by OS reconstruction from $E(4)$ covariance of the Schwinger functions (OS1). The spectral condition is part of the standard Osterwalder–Schrader reconstruction output. Core Axiom 3 is best viewed here as the upstream causal datum that is intended to be compatible with that Lorentzian interpretation; see Remark **??** below.

W1: Covariance of fields. The field operators transform covariantly under the Poincaré representation: $U(a,\Lambda)\phi(x)U(a,\Lambda)^{-1} = S(\Lambda)\phi(\Lambda x + a)$. This is inherited from the $E(4)$ invariance of the action density in the flat sector (Core Axiom 4, whose local density $\mathcal{L}(\Phi, D\Phi, \delta)$ depends on the flat metric and is therefore $E(4)$-invariant) via OS1 and analytic continuation. The spin/statistics content comes from the branch choice on Axis B.

W2: Fields as operator-valued tempered distributions. This is exactly OS0, which follows from Core Axiom 2 (finite resolution). The polynomial bounds in Schwartz seminorms, equation (**??**), guarantee that the Schwinger functions define tempered distributions.

W3: Local commutativity (microscopic causality). For spacelike separated points, $[\phi(x), \phi(y)]_\pm = 0$. This combines Core Axiom 1 (locality of the algebra) with Core Axiom 3 (finite propagation defining the causal structure). After OS reconstruction and Wick rotation, Euclidean locality becomes Minkowski locality with the light cone defined by c_{info}. The $\pm$ grading is fixed by Axis B.

W4: Cyclicity of the vacuum. The set of vectors $\{\phi(f_1)\cdots\phi(f_n)|\Omega\rangle\}$ is dense in $\mathcal{H}$. This is automatic from the GNS construction: $|\Omega\rangle$ is cyclic for $\pi(\mathcal{A})$ by definition.

Wightman axiom	Primary source	Additional hypotheses
W0 (Hilbert space, vacuum, Poincaré, spectral cond.)	Core 5 (GNS) + OS reconstruction	OS reconstruction, mixing / clustering input
W1 (covariance)	Core 4 ($E(4)$-inv. $\mathcal{L}$) + Axis B	OS reconstruction
W2 (temperedness)	Core 2	None
W3 (locality)	Core 1 + Core 3	Axis B for grading
W4 (cyclicity)	Core 5	Automatic from GNS

Table 1: Wightman axioms and their upstream sources in the present framework.

2.2 Osterwalder–Schrader Axioms

The OS axioms [?, ?] take as primitive the Schwinger functions $S_n(x_1, \ldots, x_n)$ on $(\mathbb{R}^4)^n$ and impose temperedness (OS0), Euclidean covariance (OS1), reflection positivity (OS2), clustering (OS3), and graded symmetry (OS4). In the present framework the Schwinger functions are not primitive: they are produced by the constructive dynamical system.

OS0 (Temperedness). Core Axiom 2 gives polynomial bounds on smeared observables, equation (**??**). These directly imply that the Schwinger functions define tempered distributions on $(\mathbb{R}^4)^n$. No additional hypotheses are needed.

OS1 (Euclidean covariance). Core Axiom 4 provides a local generating structure whose density, in the flat sector, depends on fields, their derivatives, and the flat metric $\delta_{\mu\nu}$. Since δ is $E(4)$-invariant, the Boltzmann weight e^{-S} is $E(4)$-invariant, and so is the equilibrium measure π. Hence all expectations under π — the Schwinger functions — are $E(4)$-invariant.

Additional hypothesis: the equilibrium measure must be uniquely determined as an $E(4)$-invariant measure. If there are multiple equilibrium measures (as in spontaneous symmetry breaking), one must select an $E(4)$-invariant one.

OS2 (Reflection positivity). This is the most complex derivation. Core Axiom 4 supplies the additive decomposition $S = S_+ + S_- + B$ across the reflection hyperplane (Lemma **??**). This lets the functional integral factorize across the hyperplane. Core Axiom 5 gives a positivity-preserving semigroup e^{-tH} with $H \geq 0$. The transfer-matrix identification (Proposition **??**) then gives $\langle \Theta F, F \rangle_E = \|e^{-t_{\min} H} F\|^2 \geq 0$.

Additional hypotheses: the transfer-matrix hypotheses of Proposition **??** — that the Euclidean measure factorizes in the right way across the hyperplane. This is supplementary analytic input and is where the constructive hard work happens in concrete theories.

Core Axiom 4 (additive decomposition) is structurally indispensable here. Without it, the transfer-matrix argument has no starting point.

OS3 (Clustering). The theorem-level hypothesis needed for OS3 in the abstract framework is the translation-mixing estimate

$$\left| \langle F\, \tau_a G \rangle_\pi - \langle F \rangle_\pi \langle G \rangle_\pi \right| \xrightarrow{|a| \to \infty} 0 \tag{1}$$

for the translation-stable observable algebra from which the Schwinger functions are built. Proposition **??** below shows that this estimate is exactly what is needed to derive OS3.

In concrete theories one often tries to obtain (**??**) from stronger dynamical input such as a spectral gap for an appropriate Euclidean transfer operator or Markov generator. That route is model-dependent and therefore lies outside the abstract theorem. The tools used there are probabilistic — Foster–Lyapunov drift conditions, coupling arguments, log-Sobolev inequalities, Bakry–Émery criteria, hypocoercivity — rather than the Euclidean field-theory tools traditionally used for clustering.

Axis D does not affect the *existence* of OS3. It refines the *rate*: under D1 (gapped), exponential clustering $S^c \sim e^{-\Delta|a|}$; under D2 (critical), polynomial clustering $S^c \sim |a|^{-2\Delta_\phi}$.

OS4 (Graded symmetry). Purely determined by the branch choice on Axis B. B1 gives full symmetry, B2 gives fermionic antisymmetry, B3 gives the mixed graded-symmetry rule, B4 preserves B3's grading with additional boson–fermion pairing constraints. No additional hypotheses.

OS axiom	Primary source	Additional hypotheses
OS0 (temperedness)	Core 2	None
OS1 (Euclidean cov.)	Core 4 (flat sector)	Unique $E(4)$-inv. equilibrium
OS2 (reflection pos.)	Core 4 + Core 5	Transfer-matrix hypotheses
OS3 (clustering)	Core 5 + Prop. **??**	Translation-mixing estimate (**??**)
OS4 (graded symmetry)	Axis B	None

Table 2: OS axioms and their upstream sources.

2.3 Haag–Kastler / Locally Covariant AQFT

The Haag–Kastler axioms [**?**] and their locally covariant generalization [**?**] take as primitive a net or functor assigning local $*$-algebras to spacetime regions or spacetimes. In the present framework this is the most natural specialization because Core Axiom 1 is already stated in Haag–Kastler language.

HK1 (Isotony). $O_1 \subset O_2$ implies $\mathcal{A}(O_1) \subset \mathcal{A}(O_2)$. This is literally the first clause of Core Axiom 1. No additional hypotheses.

HK2 (Locality / Einstein causality). For spacelike separated O_1, O_2, the algebras commute (with appropriate grading from Axis B). This is the second clause of Core Axiom 1, with causal disjointness defined by the Lorentzian metric in the manifold branch. Core Axiom 3 ensures that the dynamical notion of causality is compatible with the metric notion.

HK3 (Covariance / local covariance). In the Brunetti–Fredenhagen–Verch formulation, the primitive is a functor from the category of globally hyperbolic spacetimes to $*$-algebras with injective $*$-homomorphisms. This comes from Core Axiom 4 together with the geometric naturality stated in Remark **??**: because $\mathcal{L}$ depends locally on Φ, $D\Phi$, and g, an isometric embedding $\psi : M_1 \to M_2$ induces a pullback of field configurations and hence a pushforward of observables, giving the functorial structure.

HK4 (Time-slice property). If O contains a Cauchy surface for a larger region O', then $\mathcal{A}(O) = \mathcal{A}(O')$. This is *not* automatic from the core axioms. It requires well-posed hyperbolic evolution: the field equations must be normally hyperbolic or Dirac-type.

Core Axiom 3 is necessary but not sufficient: finite propagation ensures that information *stays within* the domain of dependence, but time-slice additionally requires that it *fills* the domain of dependence. Finite propagation prevents leaking out; hyperbolicity ensures filling in.

HK5 (Spectrum condition / positivity). This comes from Core Axiom 5 ($H \geq 0$) together with Core Axiom 3 (finite propagation → bounded dispersion → spectral support in forward cone).

Microlocal spectrum condition. On curved spacetimes, the spectrum condition is replaced by the Hadamard condition on the two-point function. This requires the additional hypothesis that the physical states are Hadamard — the curved-spacetime analog of the transfer-matrix hypothesis.

Physical algebra selection. Which algebra counts as physical depends on Axis A: A1 or A3 (gauge) gives the invariant subalgebra $\mathcal{A}_M^G(O)$ or $\mathcal{A}_M^{G_{\mathrm{disc}}}(O)$; A2 (ungauged) gives the full algebra $\mathcal{A}_M(O)$ with global symmetries acting as automorphisms.

HK/LC property	Primary source	Additional hypotheses
Isotony	Core 1 (clause i)	None
Locality	Core 1 (clause ii) + Core 3	Axis B for grading
Covariance / functoriality	Core 4 (local geom. $\mathcal{L}$)	Geometric naturality
Time-slice	Core 3 (necessary)	Hyperbolicity of field eqs.
Spectrum condition	Core 5 + Core 3	None (flat sector)
Microlocal spectrum cond.	Core 5 + Core 3	Hadamard state hyp.
Physical algebra	Axis A	None

Table 3: Haag–Kastler / locally covariant AQFT properties and their upstream sources.

2.4 Summary: What the Core Axioms Provide

The five core axioms are upstream of all three frameworks simultaneously:

- Core 1 → algebraic infrastructure (nets, isotony, locality) → shared by HK, implied by OS;
- Core 2 → regularity (temperedness, polynomial bounds) → OS0, W2;
- Core 3 → causal structure → supplies the intended upstream causal input for the Euclidean/Lorentzian correspondence;
- Core 4 → dynamical generation (additive decomposition) → hyperplane splitting for OS2; together with Remark **??** it gives functoriality for HK/LC;
- Core 5 → positivity (GNS, vacuum, bounded-below energy) → W0, OS2, HK spectrum condition.

The additional hypotheses are cleanly separated:

- Transfer-matrix hypotheses (supplementary analytic input needed for OS2);
- Translation-mixing hypothesis (**??**) (needed for OS3);
- Geometric naturality of the generating structure (needed for local covariance);
- Hyperbolicity of field equations (needed for time-slice);
- Hadamard state condition (needed for microlocal spectrum condition);
- $E(4)$-invariant equilibrium selection (needed for OS1).

This is the organizational value of the framework: in the existing literature, these ingredients are mixed together. The Wightman axioms assume the Hilbert space and spectral condition. The OS axioms assume temperedness and reflection positivity. The present framework isolates which constructive input produces which output property and clearly marks where the hard analytic work still lives.

2.5 Verification Input/Output Table

3 The Five Core Axioms

3.1 Geometric Setting

Remark 3.1 (Two geometric branches). The axioms are stated on an oriented manifold (M, g) with metric of appropriate signature. In the Euclidean branch used for Schwinger functions, g is Riemannian and the flat specialization is $(\mathbb{R}^4, \delta)$. In the Lorentzian branch

Framework	Native setting	Primitive objects	Status here
Wightman	Minkowski $\mathbb{R}^{1,3}$	Operator-valued fields, Hilbert space, Poincaré representation	Derived (flat sector)
Osterwalder–Schrader	Euclidean $\mathbb{R}^4$	Schwinger functions, reflection positivity	Derived (flat sector)
Haag–Kastler / LC-AQFT	General spacetime	Nets / functors of local $*$-algebras	Derived (manifold sector)
Present framework	Manifold-native	Core structural properties + branch data	Primitive

Table 4: The four axiomatic layers and their status in the present framework.

used for the Haag–Kastler / LC-AQFT specialization, g is Lorentzian and globally hyperbolic.

In the Lorentzian branch, causal disjointness is the usual metric notion. In the Euclidean branch there is no metric light cone; causal disjointness is defined operationally by Core Axiom 3 relative to the constructive dynamics. The locality clause of Core Axiom 1 is therefore logically downstream of Core Axiom 3 in the Euclidean branch.

3.2 Why a Hierarchical Axiom System

A scalar theory such as $\lambda\phi^4$ and a gauge theory such as Yang–Mills should agree on locality, finite resolution, finite propagation, local dynamics, and positivity, but differ on the symmetry axis. A free scalar and a free Dirac field should agree on the core axioms and differ only on the statistics axis. A massive phase and its tuned critical counterpart should agree on the same core axioms and branch choices on A, B, C, but differ on the infrared spectral axis D.

The logical structure is: core $\longrightarrow$ specialization by branch choice.

3.3 Core Axiom 1: Geometric Locality

Axiom 3.2 (Geometric Locality)**.** For each oriented manifold (M, g) with metric of appropriate signature, there exists a net of local observable $*$-algebras $\mathcal{A}_M(O)$ indexed by open regions $O \subset M$, satisfying:

(i) **Isotony**: if $O_1 \subset O_2$, then $\mathcal{A}_M(O_1) \subset \mathcal{A}_M(O_2)$;

(ii) **Locality**: if O_1 and O_2 are causally disjoint, then $[\mathcal{A}_M(O_1), \mathcal{A}_M(O_2)]_\pm = 0$.

Each $\mathcal{A}_M(O)$ is a unital $*$-algebra. In the Lorentzian branch it admits a C^*-completion. In the Euclidean branch it is represented as a unital $*$-subalgebra of $L^\infty(\pi)$ for the equilibrium measure π.

Target	Structural input	Supplementary package(s)	Imported standard step	Output
OS	Core Axioms 1–5, Axes A/B/C, flat sector	RP for OS2, CL for OS3, $E(4)$-invariant equilibrium for OS1	None	OS0–OS4
Wightman	Verified OS package above	No new package beyond RP+CL	OS reconstruction	W0–W4
HK / LC-AQFT	Core Axioms 1–5, Axis A, naturality	TS for time-slice, H for microlocal spectrum	Standard hyperbolic and Hadamard machinery	Isotony, locality, covariance, optional time-slice and microlocal spectrum
Mass gap	Verified OS package above, centered positive-time physical algebra	MG	Spectral theorem for reconstructed H	$\mathrm{spec}(H) \cap (0, \lambda_*) = \varnothing$

Table 5: Input/output table for the reusable theorem API. Structural inputs are distinguished from supplementary analytic packages and from imported standard results.

3.4 Core Axiom 2: Finite Resolution

Axiom 3.3 (Finite Resolution)**.** There exists a strictly positive resolution scale $\ell_L > 0$ such that no operationally distinguishable states resolve below ℓ_L. Smeared observables

$$O_f = \int f(x)\, O(x)\, d\,\mathrm{vol}_g(x), \qquad f \in C_c^\infty(M), \tag{2}$$

are well-defined. In the flat Euclidean sector $(M, g) = (\mathbb{R}^4, \delta)$ they satisfy polynomial bounds

$$|\langle O_{f_1} \cdots O_{f_n} \rangle| \leq C^n \prod_{k=1}^{n} \|f_k\|_{n_k}, \tag{3}$$

uniform in ℓ_L, where $\|\cdot\|_n$ denotes a Schwartz seminorm.

The axiom is structural. It says the theory is formulated at finite operational resolution. It is deliberately silent on what scaling limit, if any, is taken along a scale family.

3.5 Core Axiom 3: Finite Propagation

Axiom 3.4 (Finite Propagation)**.** There exists a maximum information speed $c_{\mathrm{info}} > 0$ such that causal influence is restricted to the domain of dependence determined by c_{info}. For any perturbation localized in a region O at time t_0, the resulting change in observables at time $t > t_0$ is supported inside the $c_{\mathrm{info}}(t - t_0)$-neighborhood of O.

Remark 3.5 (Dynamics versus equilibrium correlations). This axiom concerns the *dynamics* of the constructive system, not the *equilibrium correlations* (Schwinger functions). A nearest-neighbor Ising model has finite-range dynamics, but the equilibrium two-point function $\langle \sigma_x \sigma_y \rangle$ extends to all distances via exponential decay. No contradiction: the equilibrium measure integrates over all histories of the dynamics, including arbitrarily long chains of nearest-neighbor interactions.

Remark 3.6 (Interpretive role of finite propagation). Finite propagation is the framework's intended upstream causal datum. In the Euclidean branch, where the metric has no light cone, it gives operational content to causal disjointness and therefore to the locality clause of Core Axiom 1. In concrete constructions one expects this dynamical notion of finite information speed to be compatible with the Lorentzian causal structure appearing after Osterwalder–Schrader reconstruction.

This remark is interpretive rather than theorem-level. The actual derivation of the Wightman spectral condition in the paper comes from the OS reconstruction theorem once OS0–OS4 have been established, not from Core Axiom 3 alone.

Remark 3.7 (Causality is imposed at the dynamical level). Finite propagation is a property of the generating dynamics themselves. Causality is not an abstract requirement imposed after the fact on correlation functions; it is part of the framework at the level of information propagation.

3.6 Core Axiom 4: Local Dynamics with Additive Decomposition

Axiom 3.8 (Local Dynamics with Additive Decomposition)**.** The dynamics are generated by a local functional that decomposes additively over regions. Concretely, there exists a functional S — or, more generally, a local generating structure for the Euclidean measure or evolution law — such that for disjoint subregions $O_1, O_2 \subset M$,

$$S[\Phi|_{O_1 \cup O_2}] = S[\Phi|_{O_1}] + S[\Phi|_{O_2}] + B[\Phi|_{\partial}], \tag{4}$$

where B depends only on interface data. The local algebra $\mathcal{A}_M(O)$ is generated by observables supported in O.

Remark 3.9 (The standard Lagrangian case). In most constructive QFTs — scalar, gauge, fermionic, supersymmetric — the local generating structure is a classical action functional

$$S = \int_M \mathcal{L}(\Phi, D\Phi, g)\, d\,\mathrm{vol}_g \tag{5}$$

built from a Lagrangian density. For these, the axiom reduces to the standard statement that the action is the spacetime integral of a local density.

Remark 3.10 (General local generators). The axiom is stated more broadly than the classical Lagrangian case. It allows any local generating structure whose restriction to separated regions obeys the additive decomposition (**??**). The point of the axiom is locality of generation, not commitment to a single formalism.

Remark 3.11 (Geometric naturality for the manifold branch). The locally covariant specialization requires, in addition to additivity, naturality under orientation-preserving isometric embeddings. Concretely, if $\psi : M_1 \hookrightarrow M_2$ is such an embedding, then pullback of fields by ψ must intertwine the generating structures, and the induced pushforward of observables must define injective $*$-homomorphisms

$$\alpha_\psi : \mathcal{A}_{M_1}(O) \to \mathcal{A}_{M_2}(\psi(O))$$

satisfying $\alpha_{\psi_2 \circ \psi_1} = \alpha_{\psi_2} \circ \alpha_{\psi_1}$ and $\alpha_{\mathrm{id}_M} = \mathrm{id}_{\mathcal{A}_M}$. This is the additional geometric input used in the manifold-covariant specialization theorem.

Remark 3.12 (Scope exclusions). Theories without any local generating structure — self-dual field theories lacking even an auxiliary action formulation (such as the chiral 2-form on a 6-manifold), or theories defined purely through algebraic or functorial data without a constructive Euclidean formulation — are outside the framework's scope. This is a deliberate restriction: the transfer-matrix route to OS2 requires additive decomposition as a structural input, and any theory that lacks such decomposition cannot enter the flat-sector specialization through the present architecture. This parallels the TQFT scope boundary of Section **??**: the exclusion is stated openly.

Theories that *have* a classical action but face difficulties in defining the path integral measure — such as chiral gauge theories with anomalies — are not excluded by Core Axiom 4. Their difficulties lie elsewhere: in the well-definedness of the gauge-invariant integration and hence in the verification of Axis A1, not in the locality of the generating structure.

3.7 Core Axiom 5: Positivity and Stability

Axiom 3.13 (Positivity and Stability)**.** There exists a positivity-preserving semigroup e^{-tH} on the algebra designated as physical by the chosen symmetry branch on Axis A, with self-adjoint generator $H \geq 0$. Together with a chosen equilibrium state ω, this yields a GNS Hilbert-space representation $(\mathcal{H}, \pi, |\Omega\rangle)$ with $H|\Omega\rangle = 0$.

The core requires bounded-below energy and positivity of Euclidean time evolution. It does not require a spectral gap. Whether the selected phase is gapped or critical belongs to Axis D.

3.8 Finite Resolution and Scaling

Core Axiom 2 remains a core axiom rather than a branch condition. The framework is formulated at strictly positive operative resolution $\ell_L > 0$, and questions about continuum limits, critical limits, or scale families are separate analytic questions layered on top of that structural starting point.

If one studies a scale-labelled family of theories $\{\mathcal{T}_\lambda\}_{\lambda \in \Lambda}$, each member is evaluated against the same core axioms and branch logic. The renormalization group then compares different scales inside such a family without changing the logical status of the axioms themselves.

Thus finite resolution is not a disposable regulator inside the framework. It is the structural datum from which one may later ask whether any additional scaling theorem exists.

4 The Specialization Axes

A theory satisfying the core axioms becomes a concrete branch of constructive QFT only after branch choices are specified. Axes A–C classify field content and interaction structure. Axis D classifies the infrared phase.

4.1 Axis A: Internal Symmetry

Axiom 4.1 (A1: Gauge-Invariant Physical Algebra)**.** There exists a compact gauge group G acting locally on fields. The physical observable algebra is the gauge-invariant subalgebra $\mathcal{A}_M^{\mathrm{phys}}(O) = \mathcal{A}_M(O)^G$.

Scope. Yang–Mills with $G = SU(N)$, electrodynamics, electroweak theory, Standard Model gauge sector.

Axiom 4.2 (A2: Global Symmetry / Ungauged Theory)**.** The theory carries a global symmetry group G_{glob} (possibly trivial) but no local gauge redundancy: $\mathcal{A}_M^{\mathrm{phys}}(O) = \mathcal{A}_M(O)$.

Scope. $\lambda\phi^4$, Ising field theory, $O(N)$ models, sigma models, free theories without gauge fields.

Axiom 4.3 (A3: Discrete Gauge Theory)**.** There exists a finite gauge group G_{disc} acting locally on fields: $\mathcal{A}_M^{\mathrm{phys}}(O) = \mathcal{A}_M(O)^{G_{\mathrm{disc}}}$.

Scope. $\mathbb{Z}_N$ gauge theories, Dijkgraaf–Witten type theories.

4.2 Axis B: Statistics

Axiom 4.4 (B1: Bosonic Statistics)**.** The field algebra carries a trivial $\mathbb{Z}_2$ grading. Schwinger functions are symmetric under permutation.

Axiom 4.5 (B2: Fermionic Sector)**.** There exists a spin structure and a $\mathbb{Z}_2$-graded field algebra with commutation and anticommutation at spacelike separation according to grading. The action contains a Dirac-type kinetic term.

Axiom 4.6 (B3: Mixed Bose–Fermi Statistics)**.** The field algebra decomposes as a graded product $\mathcal{A} \sim \mathcal{A}_{\mathrm{bos}} \otimes \widehat{\mathcal{A}}_{\mathrm{ferm}}$ with combined $\mathbb{Z}_2$ grading and Yukawa-type couplings.

Axiom 4.7 (B4: Supersymmetric Mixed Sector)**.** The theory satisfies Branch B3 and, in addition, carries an odd supercharge Q such that

$$H = Q^\dagger Q \tag{6}$$

on the physical Hilbert space.

Proposition 4.8. *Under Branch B4, the positivity $H \geq 0$ of Core Axiom 5 holds automatically on the common invariant core of* (**??**)*.*

Proof. Let ψ lie in the common invariant core of Q and $Q^\dagger$. Using (**??**),

$$\langle \psi, H\psi \rangle = \langle \psi, Q^\dagger Q\psi \rangle = \langle Q\psi, Q\psi \rangle = \|Q\psi\|^2 \geq 0.$$

Hence the quadratic form of H is nonnegative on the stated core, which is exactly the positivity required in Core Axiom **??**. □

Remark 4.9. Supersymmetry is not a new axis independent of statistics; it presupposes coexistence of bosonic and fermionic sectors and then adds an algebraic relation linking them. It is therefore a refinement of B3 rather than a fifth axis.

Remark 4.10 (Protected vacuum energy and the Witten index). Under Branch B4, the quantity $\mathrm{Tr}(-1)^F e^{-\beta H}$ is stable under continuous supersymmetry-preserving deformations [?]. A nonzero index implies unbroken supersymmetry and exactly zero vacuum energy.

4.3 Axis C: Interaction Type

Axiom 4.11 (C1: Interacting Theory). At least one gauge-invariant coupling constant, self-coupling, or curvature observable is nonzero, or equivalently there exists $n \geq 4$ such that $S_n^c \not\equiv 0$.

Axiom 4.12 (C2: Free Theory). The action is quadratic: $S = \int \frac{1}{2}\Phi \cdot D \cdot \Phi \, d\,\mathrm{vol}_g$. All connected n-point functions vanish for $n \geq 3$.

4.4 Axis D: Spectral Type

Axiom 4.13 (D1: Massive / Gapped Phase). In the GNS representation, $\mathrm{spec}(H) \cap (0, \Delta) = \emptyset$ for some $\Delta > 0$. When the corresponding framework hypotheses furnish the corresponding clustering estimate, connected correlations decay exponentially.

Axiom 4.14 (D2: Critical / Conformal Phase). The vacuum sector is gapless: $\inf \mathrm{spec}\, H|_{\mathcal{H} \ominus \mathbb{C}|\Omega\rangle} = 0$. The long-distance sector is scale-invariant, and connected correlations decay polynomially.

Remark 4.15. A fixed positive ℓ_L breaks exact microscopic dilation invariance. What becomes scale-invariant at criticality is the long-distance sector, not the microscopic theory. Core Axiom 2 remains intact.

Remark 4.16 (Scope of Axis D). The present framework isolates the two clean spectral regimes most relevant to the constructive program. Gapless but nonconformal theories (e.g., theories with massless gauge bosons) motivate a future Branch D3.

4.5 Dependency Diagram and Classification Table

5 Specialization Theorems

5.1 Flat Stationary Sector

Definition 5.1. The flat stationary sector is the Euclidean specialization $(M, g) = (\mathbb{R}^4, \delta)$ with flat Euclidean metric, time-translation invariant dynamics, and Euclidean time reflection $\Theta : (x_0, \mathbf{x}) \mapsto (-x_0, \mathbf{x})$. The positive-time physical algebra $\mathcal{A}_+^{\mathrm{phys}}$ is the norm closure of bounded physical observables supported in $\{x_0 \geq \varepsilon\}$ for some $\varepsilon > 0$:

$$\mathcal{A}_+^{\mathrm{phys}} := \overline{\bigcup_{\varepsilon > 0} \mathcal{A}^{\mathrm{phys}}(\{x_0 \geq \varepsilon\})}^{\|\cdot\|_\infty}.$$

In the gauge branches, "physical" means gauge-invariant at the definition level; in the ungauged branch it means the full local algebra.

Theory / phase	A	B	C	D
$\lambda\phi^4$ massive phase	A2	B1	C1	D1
Free massive scalar	A2	B1	C2	D1
Pure Yang–Mills with mass gap	A1	B1	C1	D1
Massive free Dirac field	A2	B2	C2	D1
Wess–Zumino (unbroken massive)	A2	B4	C1	D1
$O(N)$ sigma model massive	A2	B1	C1	D1
$\mathbb{Z}_2$ gauge theory	A3	B1	C1	D1
Critical Ising / scalar fixed point	A2	B1	C1	D2
Critical $O(N)$ fixed point	A2	B1	C1	D2

Table 6: Representative theories and phases in the framework.

The centered positive-time physical algebra is

$$\mathcal{A}_+^{\mathrm{phys},0} := \{F \in \mathcal{A}_+^{\mathrm{phys}} : \langle F\rangle_\pi = 0\}.$$

Reflection maps $\mathcal{A}_+^{\mathrm{phys}}$ anti-linearly onto the corresponding negative-time algebra, spatial translations preserve $\mathcal{A}_+^{\mathrm{phys}}$, and positive Euclidean time translations preserve $\mathcal{A}_+^{\mathrm{phys}}$.

Unbounded smeared fields are not included in the definition of $\mathcal{A}_+^{\mathrm{phys}}$ itself; they are handled after reconstruction as operators affiliated with the reconstructed Hilbert-space representation.

5.2 Hyperplane Decomposition and Transfer Matrix

Lemma 5.2 (Hyperplane decomposition)**.** *In the flat Euclidean sector, the local generating functional of Core Axiom* **??** *decomposes relative to the reflection hyperplane* $\{x_0 = 0\}$*:*

$$S_E[\Phi] = S_E[\Phi_+] + S_E[\Phi_-] + B[\Phi_0], \tag{7}$$

where $\Phi_\pm$ *denotes the restriction to* $\{\pm x_0 > 0\}$ *and* B *depends only on boundary data at* $x_0 = 0$*.*

Proof. Let $O_+^\varepsilon = \{x_0 > \varepsilon\}$ and $O_-^\varepsilon = \{x_0 < -\varepsilon\}$ for $\varepsilon > 0$. These are disjoint open regions, so Core Axiom **??** gives

$$S_E[\Phi|_{O_+^\varepsilon \cup O_-^\varepsilon}] = S_E[\Phi|_{O_+^\varepsilon}] + S_E[\Phi|_{O_-^\varepsilon}] + B_\varepsilon[\Phi|_{\partial O_+^\varepsilon \cup \partial O_-^\varepsilon}].$$

Because the generating structure is local and depends on finitely many derivatives, the only couplings omitted from the two bulk terms are those crossing the slabs $|x_0| \le \varepsilon$. As $\varepsilon \downarrow 0$, these couplings collapse onto the interface $x_0 = 0$ and therefore depend only on the boundary jet Φ_0 there. Absorbing the limit into a single interface functional $B[\Phi_0]$ yields

$$S_E[\Phi] = S_E[\Phi_+] + S_E[\Phi_-] + B[\Phi_0],$$

which is exactly (**??**). □

Proposition 5.3 (Transfer-matrix factorization)**.** *Assume:*

(i) the generating functional satisfies the hyperplane decomposition of Lemma **??***;*
(ii) the Euclidean theory satisfies the standard transfer-matrix hypotheses used in constructive QFT: a reflection-invariant, hyperplane-factorizing Euclidean measure associated with S_E*;*
(iii) Euclidean time evolution is implemented by the positivity-preserving semigroup of Core Axiom **??***.*

Then for any $F \in \mathcal{A}_+^{\rm phys}$ *supported at Euclidean times* $t > t_{\rm min} > 0$*,*

$$\langle \Theta F, F\rangle_E = \langle F|e^{-2t_{\rm min}H}|F\rangle_{\mathcal{H}}. \tag{8}$$

Proof. Split the Euclidean field into negative-time, boundary, and positive-time parts: $\Phi = (\Phi_-, \Phi_0, \Phi_+)$. By Lemma **??** and the factorization hypothesis in (ii), the Euclidean measure can be written in the standard reflected form

$$d\mu_E(\Phi) = Z^{-1}\, d\mu_-(\Phi_- \mid \Phi_0)\, e^{-B[\Phi_0]}\, d\nu(\Phi_0)\, d\mu_+(\Phi_+ \mid \Phi_0),$$

where μ_- is the reflection of μ_+.

Because F is supported in $x_0 > t_{\rm min}$, only the positive-time half-space enters F. Define the boundary-to-bulk map

$$(K_{t_{\rm min}}F)(\Phi_0) := \int F(\Phi_+)\, d\mu_+^{t_{\rm min}}(\Phi_+ \mid \Phi_0),$$

where $\mu_+^{t_{\rm min}}$ is the positive-time conditional measure truncated at distance $t_{\rm min}$ from the reflection hyperplane. Reflection invariance then gives

$$\langle \Theta F, F\rangle_E = \int \overline{K_{t_{\rm min}}F(\Phi_0)}\, K_{t_{\rm min}}F(\Phi_0)\, e^{-B[\Phi_0]} d\nu(\Phi_0) = \|K_{t_{\rm min}}F\|^2_{L^2(\nu_B)},$$

where $d\nu_B := e^{-B[\Phi_0]}d\nu(\Phi_0)$.

The standard Osterwalder–Schrader transfer-operator construction identifies $K^*_{t_{\rm min}}K_{t_{\rm min}}$ with the transfer operator $T_{2t_{\rm min}}$. Hypothesis (iii) states that this transfer operator is implemented on the GNS Hilbert space by the positivity-preserving semigroup e^{-tH}. Therefore

$$\langle \Theta F, F\rangle_E = \langle F, T_{2t_{\rm min}}F\rangle_{\mathcal{H}} = \langle F, e^{-2t_{\rm min}H}F\rangle_{\mathcal{H}},$$

which is the claimed identity. □

Remark 5.4 (Status of OS2). The framework isolates the structural ingredients behind reflection positivity but does not claim that every local generating structure automatically yields a transfer matrix. The transfer-matrix hypotheses remain supplementary analytic input.

5.3 The Upstream Dynamical Chain for OS3

The abstract framework isolates the exact clustering hypothesis needed for OS3 and keeps model-dependent dynamical routes to that hypothesis separate.

Proposition 5.5 (Translation mixing implies OS3)**.** *Let* π *be an* $E(4)$*-invariant stationary Euclidean measure in the flat sector, and let* $\mathcal{B} \subset L^\infty(\pi)$ *be a unital algebra of observables that is closed under translations:* $\tau_a\mathcal{B} \subset \mathcal{B}$ *for all* $a \in \mathbb{R}^4$*. Assume that the*

Schwinger functions are obtained as expectations of finite products of observables from $\mathcal{B}$, and that for all $F, G \in \mathcal{B}$,

$$\big|\langle F\,\tau_a G\rangle_\pi - \langle F\rangle_\pi \langle G\rangle_\pi\big| \xrightarrow{|a|\to\infty} 0. \tag{9}$$

Then the associated Schwinger functions satisfy OS3.

Proof. Let

$$F = O_{f_1}\cdots O_{f_m}, \qquad G = O_{g_1}\cdots O_{g_n},$$

with each factor in $\mathcal{B}$. Since $\mathcal{B}$ is an algebra and is translation-stable, we have $F, G, \tau_a G \in \mathcal{B}$. By definition of the Schwinger functions,

$$\langle F\,\tau_a G\rangle_\pi = S_{m+n}(f_1,\dots,f_m,\tau_a g_1,\dots,\tau_a g_n),$$

while

$$\langle F\rangle_\pi = S_m(f_1,\dots,f_m), \qquad \langle G\rangle_\pi = S_n(g_1,\dots,g_n).$$

Applying (**??**) therefore gives

$$S_{m+n}(f_1,\dots,f_m,\tau_a g_1,\dots,\tau_a g_n) \xrightarrow{|a|\to\infty} S_m(f_1,\dots,f_m)\,S_n(g_1,\dots,g_n),$$

which is exactly the Osterwalder–Schrader clustering axiom OS3 for observables generated by $\mathcal{B}$. □

Theorem 5.6 (Spectral gap route to time-direction mixing)**.** *Let the hypotheses of Proposition **??** hold, and let $\mathcal{C}_+^{\mathrm{phys},0} \subset \mathcal{A}_+^{\mathrm{phys},0}$ be a centered class of positive-time observables such that:*

(i) for all $F, G \in \mathcal{C}_+^{\mathrm{phys},0}$ and all $t \geq 0$, Euclidean time translation by t is represented by

$$\langle F\,\tau_{te_0} G\rangle_\pi - \langle F\rangle_\pi\langle G\rangle_\pi = \langle \widehat{F}, e^{-tH}\widehat{G}\rangle_{\mathcal{H}};$$

(ii) the reconstructed Hamiltonian has a spectral gap $\lambda > 0$ on $\Omega^\perp$, equivalently

$$\|e^{-tH}P_{\Omega^\perp}\| \leq e^{-\lambda t} \qquad (t \geq 0);$$

(iii) the Schwinger functions are generated by a translation-stable algebra containing $\mathcal{C}_+^{\mathrm{phys},0}$.

Then the connected correlations satisfy exponential mixing along the Euclidean time direction:

$$\big|\langle F\,\tau_{te_0} G\rangle_\pi - \langle F\rangle_\pi\langle G\rangle_\pi\big| \leq e^{-\lambda t}\|\widehat{F}\|_{\mathcal{H}}\|\widehat{G}\|_{\mathcal{H}}.$$

Proof. Since F and G are centered, $\widehat{F}, \widehat{G} \in \Omega^\perp$. By hypothesis (i),

$$\langle F\,\tau_{te_0} G\rangle_\pi - \langle F\rangle_\pi\langle G\rangle_\pi = \langle \widehat{F}, e^{-tH}P_{\Omega^\perp}\widehat{G}\rangle_{\mathcal{H}}.$$

Cauchy–Schwarz and hypothesis (ii) give

$$\big|\langle F\,\tau_{te_0} G\rangle_\pi - \langle F\rangle_\pi\langle G\rangle_\pi\big| \leq \|\widehat{F}\|_{\mathcal{H}}\|e^{-tH}P_{\Omega^\perp}\widehat{G}\|_{\mathcal{H}} \leq e^{-\lambda t}\|\widehat{F}\|_{\mathcal{H}}\|\widehat{G}\|_{\mathcal{H}}.$$

This is the required exponential mixing estimate in the time direction. □

Corollary 5.7 (Gap route to Package CL)**.** *In addition to the hypotheses of Theorem* **??***, assume there is an* $E(4)$*-stable generating algebra* $\mathcal{B}^{\rm phys}$ *such that every pair* $F, G \in \mathcal{B}^{\rm phys}$ *can, after an* $E(4)$ *rotation of coordinates, be represented by elements of* $\mathcal{C}_+^{\rm phys,0}$ *to which Theorem* **??** *applies. Then* $\mathcal{B}^{\rm phys}$ *satisfies the translation-mixing estimate* (**??**)*. Hence Package CL holds, and Proposition* **??** *yields OS3.*

Proof. By assumption, each large-translation correlation in $\mathcal{B}^{\rm phys}$ can be rotated into the time-direction situation covered by Theorem **??**. The $E(4)$-invariance of the equilibrium state transports the resulting estimate back to the original configuration, yielding (**??**). □

Remark 5.8 (Axis D refines the rate). Axis D now enters the theorem library in a precise way. Under D1, Theorem **??** together with Corollary **??** is the canonical route to exponential clustering once a spectral gap is verified on a sufficiently rich observable class. Under D2, one expects a different package replacing the exponential estimate of Theorem **??** by a polynomial decay estimate. Thus Axis D does not change the logical place where OS3 is proved; it changes the quantitative package that realizes Package CL.

5.4 Derivation of OS0–OS4

Theorem 5.9 (Flat stationary specialization — detailed form)**.** *In the flat stationary sector, a system satisfying Core Axioms* **??***–***??** *together with branch choices on Axes A, B, and C produces Schwinger functions satisfying:*

(i) OS0 from Core Axiom **??***;*

(ii) OS1 from Core Axiom **??** *in the flat Euclidean sector, under the hypothesis that the equilibrium measure is uniquely* $E(4)$*-invariant;*

(iii) OS2 from Core Axioms **??** *and* **??***, under the transfer-matrix hypotheses of Proposition* **??***;*

(iv) OS4 from the chosen branch on Axis B.

If, in addition, the translation-mixing estimate (**??**) *holds for the observable algebra generating the Schwinger functions, then OS3 holds by Proposition* **??***. OS reconstruction then yields Wightman functions satisfying W0–W4.*

Proof. **OS0.** By Core Axiom **??**, smeared observables satisfy (**??**). Therefore the Schwinger functions define continuous multilinear functionals on Schwartz test functions, hence tempered distributions on $(\mathbb{R}^4)^n$.

OS1. In the flat sector, the local generating density depends on Φ, $D\Phi$, and $\delta_{\mu\nu}$. Hence S is $E(4)$-invariant, e^{-S} is $E(4)$-invariant, and the equilibrium measure (uniquely selected by hypothesis) inherits that invariance. Expectations with respect to π are therefore $E(4)$-covariant.

OS2. For $F \in \mathcal{A}_+^{\rm phys}$, Proposition **??** gives $\langle \Theta F, F\rangle_E = \|e^{-t_{\min}H}F\|_{\mathcal{H}}^2 \geq 0$, since $H \geq 0$ by Core Axiom **??**.

OS4. The grading is determined by Axis B.

OS3. Under the translation-mixing hypothesis, Proposition **??** applies directly.

Wightman reconstruction. Once OS0–OS4 are established, the OS reconstruction theorem [?, ?] yields a Hilbert space, vacuum vector, operator-valued tempered fields, and a positive-energy representation of the Euclidean continuation, hence the Wightman axioms W0–W4. Core Axiom **??** identifies the upstream algebraic origin of locality, and Core Axiom **??** explains why causal propagation is built into the Euclidean framework before reconstruction (Remark **??**). □

5.5 Density, Separation, and Mass-Gap Transfer

Proposition 5.10 (Positive-time density after reconstruction)**.** *Assume a reflection-positive Euclidean theory for which the standard OS reconstruction has been carried out and suppose that the OS images $\{\widehat{F} : F \in \mathcal{A}_+^{\mathrm{phys}}\}$ are dense in $\mathcal{H}$. Then the images of centered positive-time observables $\{\widehat{F} : F \in \mathcal{A}_+^{\mathrm{phys},0}\}$ are dense in $\Omega^\perp$.*

Proof. Let $\psi \in \Omega^\perp$ and $\varepsilon > 0$. By density of $\{\widehat{F} : F \in \mathcal{A}_+^{\mathrm{phys}}\}$ in $\mathcal{H}$, choose $F \in \mathcal{A}_+^{\mathrm{phys}}$ with $\|\psi - \widehat{F}\|_{\mathcal{H}} < \varepsilon$. Set

$$F_0 := F - \langle F \rangle_\pi \mathbf{1}.$$

Then $F_0 \in \mathcal{A}_+^{\mathrm{phys},0}$ and

$$\widehat{F_0} = \widehat{F} - \langle \Omega, \widehat{F} \rangle_{\mathcal{H}}\, \Omega.$$

Since $\psi \perp \Omega$, orthogonal projection onto $\Omega^\perp$ cannot increase the distance to ψ, so

$$\|\psi - \widehat{F_0}\|_{\mathcal{H}} = \|P_{\Omega^\perp}(\psi - \widehat{F})\|_{\mathcal{H}} \le \|\psi - \widehat{F}\|_{\mathcal{H}} < \varepsilon.$$

Therefore the images of centered positive-time observables are dense in $\Omega^\perp$. □

Proposition 5.11 (Spectral separation by the physical algebra)**.** *Assume the set $\{\widehat{F} : F \in \mathcal{A}_+^{\mathrm{phys},0}\}$ is dense in $\Omega^\perp$.*

(i) For every nonzero vector $\psi \in \Omega^\perp$, there exists $F \in \mathcal{A}_+^{\mathrm{phys},0}$ such that $\langle \psi, \widehat{F} \rangle_{\mathcal{H}} \neq 0$.

(ii) If P_I is a nonzero spectral projection of $H|_{\Omega^\perp}$ for a Borel set $I \subset (0, \infty)$, then there exists $F \in \mathcal{A}_+^{\mathrm{phys},0}$ such that $P_I \widehat{F} \neq 0$.

Proof. If (i) failed for some nonzero $\psi \in \Omega^\perp$, then ψ would be orthogonal to a dense subset of $\Omega^\perp$ and hence to all of $\Omega^\perp$, contradicting $\psi \neq 0$. For (ii), choose any nonzero ψ in the range of P_I and apply (i). If $P_I \widehat{F} = 0$ for every F, then $\langle \psi, \widehat{F} \rangle_{\mathcal{H}} = \langle P_I \psi, \widehat{F} \rangle_{\mathcal{H}} = 0$ for every F, again contradicting (i). □

Theorem 5.12 (OS reconstruction plus gap transfer implies Minkowski mass gap)**.** *Assume:*

(i) OS0–OS4 hold, so the standard OS reconstruction yields a Hilbert space $\mathcal{H}$, vacuum vector Ω, and self-adjoint Hamiltonian $H \ge 0$;

(ii) Package MG of Definition **??** *holds.*

Then

$$\operatorname{spec}(H) \cap (0, \lambda_*) = \varnothing.$$

Proof. By MG2, for every $F \in \mathcal{C}_+^{\mathrm{phys},0}$ and $t \ge 0$,

$$\|e^{-tH} \widehat{F}\|_{\mathcal{H}} \le C_{\mathrm{MG}} e^{-\lambda_* t} \|\widehat{F}\|_{\mathcal{H}}.$$

Since e^{-tH} is bounded and MG3 gives density of $\{\widehat{F} : F \in \mathcal{C}_+^{\mathrm{phys},0}\}$ in $\Omega^\perp$, the same estimate extends by continuity to every $\psi \in \Omega^\perp$:

$$\|e^{-tH} \psi\|_{\mathcal{H}} \le C_{\mathrm{MG}} e^{-\lambda_* t} \|\psi\|_{\mathcal{H}}.$$

Suppose, for contradiction, that $\operatorname{spec}(H) \cap (0, \lambda_*) \neq \varnothing$. Then there exists $\mu < \lambda_*$ and a nonzero spectral projection $P_{[0,\mu]}$ on $\Omega^\perp$. By Proposition **??**, or directly from MG3, choose nonzero $\psi \in \operatorname{Ran} P_{[0,\mu]}$. The spectral theorem gives

$$\|e^{-tH} \psi\|_{\mathcal{H}}^2 = \int_{[0,\mu]} e^{-2tE}\, d\|P_E \psi\|^2 \ge e^{-2\mu t} \|\psi\|_{\mathcal{H}}^2.$$

Combining this with the upper bound above yields

$$e^{(\lambda_*-\mu)t}\|\psi\|_{\mathcal{H}} \le C_{\mathrm{MG}}\|\psi\|_{\mathcal{H}} \qquad (t \ge 0),$$

which is impossible as $t \to \infty$ because $\lambda_* - \mu > 0$ and $\psi \neq 0$. Therefore no spectrum can lie in $(0, \lambda_*)$. □

5.6 General Manifold Specialization

Theorem 5.13 (Manifold-covariant specialization — detailed form)**.** *On a globally hyperbolic Lorentzian manifold, a system satisfying Core Axioms* **??***–***??***, a choice on Axis A, and geometric naturality in the sense of Remark* **??** *induces a net of physical local $*$-algebras satisfying:*

(a) Isotony and locality, from Core Axioms **??** *and* **??***;*

(b) Physical-algebra selection, from Axis A;

(c) Local covariance, from Core Axiom **??** *together with Remark* **??***.*

If, in addition, the local algebra is generated by on-shell fields modulo the equations of motion and those equations are normally hyperbolic or Dirac-type, then the time-slice property holds. If the physical states are Hadamard, then the microlocal spectrum condition holds.

Proof. Isotony is exactly clause (i) of Core Axiom **??**, and locality is clause (ii), with causal disjointness computed in the Lorentzian metric and controlled dynamically by Core Axiom **??**. Axis A specifies which local algebra is physical: the invariant subalgebra in the gauge branches and the full algebra in the ungauged branch.

For local covariance, let $\psi : M_1 \hookrightarrow M_2$ be an orientation-preserving isometric embedding. By Remark **??**, pullback of fields by ψ induces an injective $*$-homomorphism

$$\alpha_\psi : \mathcal{A}_{M_1}(O) \to \mathcal{A}_{M_2}(\psi(O))$$

compatible with composition and identities. The assignment $M \mapsto \mathcal{A}_M$ together with $\psi \mapsto \alpha_\psi$ is therefore a covariant functor on the spacetime category.

If the local algebra is generated by on-shell fields modulo the equations of motion and those equations are normally hyperbolic or Dirac-type, then data on any neighborhood of a Cauchy surface determine the solution uniquely on its domain of dependence. Hence every observable in the larger region is represented by data already generated in the time-slice, so the time-slice property holds. If the physical states are Hadamard, then their two-point functions have the standard microlocal wavefront set, and the microlocal spectrum condition follows by the usual Radzikowski characterization [**?**]. □

5.7 How To Use the Framework in a Realization Paper

The intended use in a model-verification paper is the following checklist.

1. Verify the structural flat package H_{struct} of Definition **??**: Core Axioms 1–5, branch choices, the flat stationary specialization, the chosen equilibrium state, and the physical positive-time algebra of Definition **??**.

2. Verify Package RP if OS2 is needed. This unlocks Proposition **??** and therefore OS2.

3. Verify either Package CL directly, or verify the spectral-gap hypotheses of Theorem **??** and deduce Package CL. This unlocks Proposition **??** and therefore OS3.
4. Invoke Theorem **??** to conclude OS0–OS4 and then W0–W4 by standard OS reconstruction.
5. If a Minkowski mass-gap statement is desired, verify the density/separation statements via Propositions **??** and **??**, package the quantitative estimate as Package MG, and invoke Theorem **??**.
6. In the Lorentzian branch, verify naturality first, then Package TS for time-slice and Package H for the microlocal spectrum condition, and invoke Theorem **??**.

Corollary 5.14 (Verification schema for realization papers)**.** *A realization paper may be organized as:*

(i) structural verification of H_{struct};
(ii) verification of RP;
(iii) verification of CL directly or via Theorem **??***;*
(iv) invocation of Theorem **??***;*
(v) optional verification of MG and invocation of Theorem **??***.*

In the gauge branches, the same schema is carried out on the gauge-invariant physical sector using Corollary **??***.*

Proof. This is a direct restatement of the theorem library established in this section. □

6 Compatibility, Non-Redundancy, and Separating Examples

6.1 Compatibility

Remark 6.1. Compatibility is part of the architecture rather than a separate theorem: the core axioms constrain locality, resolution, propagation, generation, and positivity, while the specialization axes classify symmetry, statistics, interaction type, and infrared phase. No axis changes the meaning of a core axiom, and no core axiom fixes a branch value. The examples below record how the same core can be combined with different branch choices.

6.2 Non-Redundancy of the Specialization Axes

The following are separating examples at the intended structural level; they are not model-theoretic independence proofs and are stated as examples rather than theorems for exactly that reason.

Example 6.2 (Axis A distinguishes gauge from ungauged theories)**.** $\lambda\phi^4$ (Branch A2) and pure $SU(N)$ Yang–Mills (Branch A1) both satisfy the core axioms at the intended structural level but differ on Axis A.

Example 6.3 (Axis B distinguishes statistics sectors)**.** A free massive scalar (B1) and a free Dirac field (B2) satisfy the same core axioms and Branches A2, C2, but differ on Axis B.

Example 6.4 (Axis C distinguishes free from interacting theories)**.** The free massive scalar (C2) and $\lambda\phi^4$ (C1) satisfy the same core axioms and Branches A2, B1, but differ on Axis C.

Example 6.5 (Axis D distinguishes infrared phases)**.** A scalar theory on A2, B1, C1 with tunable parameters gives D1 (gapped) at generic parameters and D2 (critical) at a tuned critical point, with the same core axioms and A/B/C branch data.

6.3 Core Separating Examples

Remark 6.6. The following examples illustrate why no core axiom should be regarded as obviously redundant.

Example 6.7 (Locality can fail while the other structures remain)**.** A theory with a single global observable algebra and no region-by-region assignment may still have finite resolution, finite propagation, additive generation, and positivity, but it has no isotony/locality net and therefore fails Core Axiom **??**.

Example 6.8 (Finite resolution can fail while locality survives)**.** A continuum theory with uncontrolled ultraviolet singularities may still have local algebras, finite propagation, local generation, and positive energy, but it fails the polynomial temperedness bounds of Core Axiom **??**.

Example 6.9 (Finite propagation can fail)**.** Brownian motion has instantaneous spreading and therefore violates finite information speed while still serving as an example of local observables with additive generation and positivity.

Example 6.10 (Additive decomposition can fail)**.** A bilocal or mean-field action violates the separated-region additivity required in Core Axiom **??** while leaving the other structural notions meaningful.

Example 6.11 (Positivity can fail)**.** A wrong-sign kinetic term yields an energy form unbounded below even if one still writes down local observables, finite resolution data, and local equations of motion.

7 Constructiveness, Renormalization, and Scaling

7.1 Why "Constructive"

The term is used in contrast with downstream axiomatizations. The Wightman axioms ask for Hilbert-space data; the OS axioms ask for Schwinger functions; the Haag–Kastler axioms ask for nets. In all cases one verifies properties of an already-produced theory.

The present axioms ask for direct structural properties of the generating dynamical system: does it admit a net of local algebras? Does it have finite operational resolution? Does it propagate information at finite speed? Is its generating structure local with additive decomposition? Is the generator positive?

It is important to distinguish this usage from the standard meaning of "constructive" in mathematical physics (as in Glimm–Jaffe), which refers to explicit constructions with detailed analytic estimates. The present paper isolates the *structural* layer of a constructive program. The analytic layer — where one proves existence and properties of concrete theories using whatever techniques are appropriate — is logically separate from the framework itself.

7.2 What the Axioms Prove and What They Do Not

The axioms describe a constructive theory at one positive scale. A continuum family $\{\mathcal{T}_\varepsilon\}_{\varepsilon>0}$ with $\ell_L(\varepsilon) \to 0$ requires an additional limiting theorem.

The core axioms do not by themselves prove that a continuum limit exists, that a critical point exists, that a spectral gap exists, or that transfer-matrix factorization holds. They sharply separate these analytic tasks from the structural question of whether the underlying dynamical system has the right architecture.

7.3 Renormalization Group Within the Framework

Let $\mathcal{T}_{\ell_L}$ denote a theory at scale ℓ_L. Coarse-graining to $\ell' > \ell_L$ gives $\mathcal{T}_{\ell_L} \mapsto \mathcal{T}_{\ell'}$ with modified couplings but the same structural type. Each member of this scale family is evaluated against the same core axioms and branch logic.

If a theory is constructed by cutoff removal, the bare parameters may depend on the cutoff. If it is constructed at fixed ℓ_L and analyzed through a different limiting procedure, the physically relevant running concerns effective couplings extracted after coarse-graining, while the microscopic resolution stays positive.

8 Supersymmetry, Topological Theories, and the Boundary of Scope

8.1 Supersymmetry Fits the Framework Cleanly

Supersymmetry creates no friction with the five core axioms. The correct structural place is Branch B4. The key constructive gains are:

1. **Positivity becomes internally certified.** Under B4, $H = Q^\dagger Q$ gives $H \geq 0$ automatically (Proposition **??**). This reduces the verification burden for Core Axiom 5 in the supersymmetric sector.
2. **Vacuum protection.** The Witten index is stable under continuous deformations [**?**]. When nonzero, the vacuum energy vanishes exactly.
3. **Transfer-matrix strengthening.** Once Euclidean time evolution is identified with e^{-tH}, the factorization $H = Q^\dagger Q$ gives an additional internal reason that the transfer semigroup is positive.

8.2 Pure TQFTs Mark a Genuine Scope Boundary

Purely topological quantum field theories create real friction with the present axioms. The core axioms were designed for QFTs with local propagating degrees of freedom. Pure TQFTs do not have such degrees of freedom. As a result, several core axioms become vacuous or degenerate:

- **Core Axiom 1.** A TQFT can assign finite-dimensional algebras to regions, but the dependence is topological rather than metric. The axiom holds only degenerately.
- **Core Axiom 2.** Finite resolution is conceptually mismatched: polynomial bounds hold trivially because the spaces are finite-dimensional, but they control no nontrivial ultraviolet structure.
- **Core Axiom 3.** Finite propagation is vacuous when there are no propagating local excitations.

- **Core Axiom 4.** This is the least problematic: Chern–Simons and BF theories admit local action functionals.
- **Core Axiom 5.** Often $H = 0$, so the axiom holds vacuously.

Pure TQFTs lie at the boundary of scope rather than in the main body. For them, the Atiyah–Segal functorial formulation [?] is the more natural axiomatic language.

8.3 Topological Sectors Inside Dynamical Theories

The framework does capture topological sectors of dynamical theories whenever those sectors arise inside a theory with propagating local degrees of freedom. Examples: θ-vacuum structure in Yang–Mills, topological protection of the Witten index. What lies outside scope is not topological information as such, but theories that are purely topological with no propagating sector at all.

9 Discussion and Outlook

9.1 Summary of Advances

The main advances of the framework are:

1. The detailed correspondence between the five core axioms and each of the three standard axiomatic frameworks (Section ??), with explicit identification of which core axiom produces which downstream property and where additional hypotheses are needed.
2. The reformulation of Core Axiom 4 as "local dynamics with additive decomposition" rather than "local action functional," making the framework broad enough to accommodate local generators beyond the classical Lagrangian case (Remark ??).
3. The replacement of the overstrong OS3 claim by the exact theorem-level mixing hypothesis (??) together with a full proof that this hypothesis yields OS3 (Proposition ??), while keeping spectral-gap arguments in their proper model-dependent role.
4. The explanation of why Core Axiom 3 (finite propagation) is essential for the Euclidean-to-Lorentzian transition: it provides causal structure in the Euclidean branch, justifies the Wick rotation, and produces Lorentz invariance and the spectral condition in the reconstructed Lorentzian theory (Remark ??).
5. Integration of supersymmetry as Branch B4 and clear treatment of TQFTs as a scope boundary.
6. The explicit addition of geometric naturality (Remark ??) as the hypothesis that closes the gap between local generation and genuine locally covariant functoriality.
7. Positioning relative to the active constructive QFT programs of Hairer, Gubinelli–Imkeller–Perkowski, and the Balaban program.

9.2 What Is Not Claimed

This paper does not claim that the five core axioms characterize all quantum field theories, nor that the transfer-matrix, Hadamard, spectral-gap, or critical-scaling hypotheses follow automatically from them. It does not claim that every continuum limit exists, that every critical point exists, or that every perturbative series can be promoted to a full constructive theory.

The paper is a framework paper: it isolates upstream structural conditions and shows how the standard axiomatic frameworks reappear downstream once the corresponding additional analytic hypotheses are imposed.

9.3 Further Refinements

Gapless nonconformal spectral branch. Theories with massless gauge bosons suggest a future Branch D3.

Extended supersymmetry. Branch B4 captures the basic supersymmetric strengthening. Extended SUSY, twisting, and other refinements can be layered on top.

Topological sectors inside dynamical theories. The right direction is not to force pure TQFTs into the core but to analyze topological sectors (e.g., θ-vacua) inside propagating theories.

Relation to singular SPDE methods. The framework should be compatible with singular-SPDE methods such as Hairer's regularity structures [?] or paracontrolled distributions [?]. Making that compatibility fully explicit would sharpen the connection.

10 Conclusion

The present framework proposes five core axioms shared by all constructive quantum field theories with propagating local degrees of freedom, together with four specialization axes. The core axioms supply locality, finite resolution, finite propagation, local dynamics with additive decomposition, and positivity. The axes classify symmetry, statistics, interaction type, and infrared phase.

The framework's value lies not in proving new results about specific models but in organizing the logical dependencies cleanly. In the flat stationary sector, the core axioms plus branch data and supplementary analytic hypotheses yield the OS axioms and hence the Wightman axioms. In the Lorentzian manifold branch, they yield the Haag–Kastler / locally covariant AQFT structure. The places where the framework should not be stretched — such as pure TQFTs or theories without any local generating structure — are stated openly.

The paper is therefore an abstract framework statement: it proposes an organizational architecture, maps it in detail to the standard frameworks, and separates the universal structural inputs from the additional analytic hypotheses required for concrete applications.

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